Economic, Political & Policy Sciences

Data Analytics Challenge

Airline Crew Sequences Meet Bad Weather

Airlines often face disruptions caused by delays, weather, and operational constraints. Understandably, certain airports are more subject to weather delays than others, particularly during certain seasons of the year. In order to minimize these delays when they happen, efforts can be made to schedule flights more intelligently. One major piece of this challenge is pilot scheduling, which involves deciding which flights should be assigned to the same pilot in a combination called a 'sequence'. Sequences of at-risk flights can lead to cascading delays, missed connections, and increased fatigue risk.

The goal of this challenge is to identify pairs of flights that should not be part of the same pilot's sequence in order to reduce disruptions and improve reliability. In real life, these pilot sequences can span many flights over several days, but for this challenge we'll consider only an inbound flight, from airport A, to DFW, American's largest hub, and an outbound flight to airport B. A and B can be any airport in the United States.

Example:

Airports A and B are both prone to sudden thunderstorms in spring, while airport C is not. Hence, assigning a pilot to fly A to DFW to B (or vice versa) is more likely to lead to delays than assigning them to fly A to DFW to C or B to DFW to C (or vice versa). We would like to identify these pairs (A and B, in this case) that represent sub-optimal combinations so that we can avoid assigning them in future sequences.

Objective:

Find flights that should not be part of a pilot's sequence at the same time to minimize:

- Delay propagation across multiple flights
- Duty time violations
- Missed connections due to tight turnarounds
- Increased fatigue and operational risk

Expectations:

We know that you only have a couple of weeks to work on this, so we're not expecting a fully-formed and vetted model. If you have a chance to build a model that's great, but if not, we'd love for you to come to our discussion having familiarized yourself with the data sources below and ready to explain your thoughts on how you would go about building such a model. Examples of some questions we may discuss include:

- What features might be important and what other data might you need?
- What type of model would work best for this problem?
- How would you deal with the seasonality of the data?
- What issues might arise from the sparsity of severe weather events?
- What accuracy metrics would be appropriate?

Data Sources:

Flight Data

- **FAA Aviation System Performance Metrics (ASPM)**
  https://aspm.faa.gov
  Flight schedules, delays, and performance metrics.

- **Bureau of Transportation Statistics (BTS) – On-Time Performance**
  https://www.transtats.bts.gov/OT_Delay/OT_DelayCause1.asp
  Detailed flight-level delay data.

- **OpenSky Network**
  https://opensky-network.org
  Real-time and historical flight tracking data.

Weather Data

- **NOAA National Weather Service**
  https://www.weather.gov
  Historical and forecasted weather data.

- **Aviation Weather Center (AWC)**
  https://aviationweather.gov
  METAR, TAF, and other aviation-specific weather products.

- **OpenWeather API**
  https://openweathermap.org/api
  Real-time and historical weather conditions.

Submission guideline

- Each team should prepare a team report including your ideas/solutions as word or pdf document no longer than 20 pages.
- Each team should submit their team report by 5pm on April 17th (Friday) via this link.

- The top seven finalists will be notified by April 21st and invited to give a 7-min presentation of their team report during the EPPS-AA event at 2pm on April 24th (Friday).
- The 1st, 2nd and 3rd place winners will be selected after their presentations and awarded with prizes and award certificate during the event.
- For any question about the case, please contact Jose Ramirez-Hernandez, Senior Lead Data Scientist at American Airlines (Jose.Ramirez-hernandez@aa.com).